EXPLORATORY DATA ANALYSIS

E-Commerce Sales & Customer Behaviour Analysis

A data-driven review of retail sales, product demand, customer segments and geographic performance

Dataset: UCI Online Retail Dataset
Tools: Python · Pandas · NumPy · Matplotlib · Seaborn

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1. Executive Summary

This project explores the UCI Online Retail transactional dataset using the supplied prepared sales and returns extracts. The positive-sales extract contains **524,878 sales lines** and records approximately **£10,642,111** in line revenue across the period 01 Dec 2010 to 09 Dec 2011. The extract includes 19,960 distinct invoice identifiers and 4,338 identifiable customers.

Revenue is geographically concentrated: the United Kingdom contributes **84.6%** of revenue in the country summary. November 2011 is the highest-revenue month in the supplied monthly table at **£1,503,867**. RFM segmentation identifies 1,312 Champions with about £6,476,425 segment revenue.

These findings support prioritising retention of high-value customers, reviewing the UK-heavy market mix, planning inventory for seasonal demand and monitoring return/cancellation activity. They are descriptive associations, not causal conclusions.

A reconciliation note is important: the supplied KPI table reports 19,960 distinct invoices, while counting invoice identifiers in the positive-sales extract gives 19,962. The cleaning summary reports 536,641 cleaned master rows, while the supplied positive-sales and return/cancellation extracts sum to 535,465 rows. These differences are preserved and disclosed rather than silently reconciled.

2. Introduction and Objectives

The assignment asks for an EDA of sales performance, customer purchasing behaviour, product demand and geographic distribution. The analysis is structured around the assignment questions: revenue-generating products and countries, monthly sales patterns, quantity demand, average order value, valuable customers, unusual transactions, cancellations/returns, product demand and the relationship between price and quantity.

Objectives

- Understand the structure and quality of the supplied retail extracts.
- Summarise revenue, order, product and customer patterns.
- Explore distributions, relationships, time patterns and customer segments.
- Translate the strongest descriptive findings into practical recommendations.

3. Dataset Overview

The reference dataset is the UCI Online Retail Dataset, a UK-based online retailer's transactional data. The assignment specifies the UCI source. The supplied ZIP contains prepared positive-sales and return/cancellation extracts plus summary tables; it does not include the original UCI workbook. Therefore, this report analyses the supplied extracts and references the original dataset source for provenance.

Positive-sales extract: 524,878 rows × 15 columns. **Returns/cancellations extract:** 10,587 rows × 12 columns. The sales extract spans 2010-12-01 to 2011-12-09, with 38 countries and 3,922 distinct stock codes.

Important fields include InvoiceNo (invoice identifier), StockCode (product identifier), Description (product name), Quantity (units on a line), InvoiceDate (transaction timestamp), UnitPrice (price per unit in GBP), CustomerID (customer identifier), Country, LineRevenue (line-level revenue) and derived calendar fields such as YearMonth, Weekday and Hour.

4. Data Quality and Preparation

The supplied cleaning summary records 541,909 original rows, 5,268 exact duplicate rows removed and 536,641 rows in the cleaned master dataset. It reports 524,878 positive-sales lines, 10,587 return/cancellation-related lines, 132,186 sales rows with missing CustomerID and no missing Description values.

In the positive-sales extract, missing CustomerID values represent **25.2%** of sales lines. This matters because customer-level metrics and RFM segmentation can only use records with an identified customer. The analysis does

not impute missing customer identifiers.

Numeric fields and timestamps were parsed for analysis. Positive sales lines were analysed separately from the supplied return/cancellation extract to avoid treating negative quantities as ordinary demand. Outliers were investigated using distribution and box-plot views; for visual readability, histograms are capped at the 99th percentile and box-plot fliers are hidden in the displayed plots. The underlying data are not clipped in the analytical tables.

The two reconciliation differences noted in the executive summary should be checked against the original workbook and preprocessing script before a final audited financial total is claimed.

5. Exploratory Data Analysis

5.1 Univariate distributions

Sales-line quantity, unit price and line revenue are strongly right-skewed. A small number of unusually large values stretch the distributions, so the report uses percentile-capped histograms for readability while retaining original values for calculations.

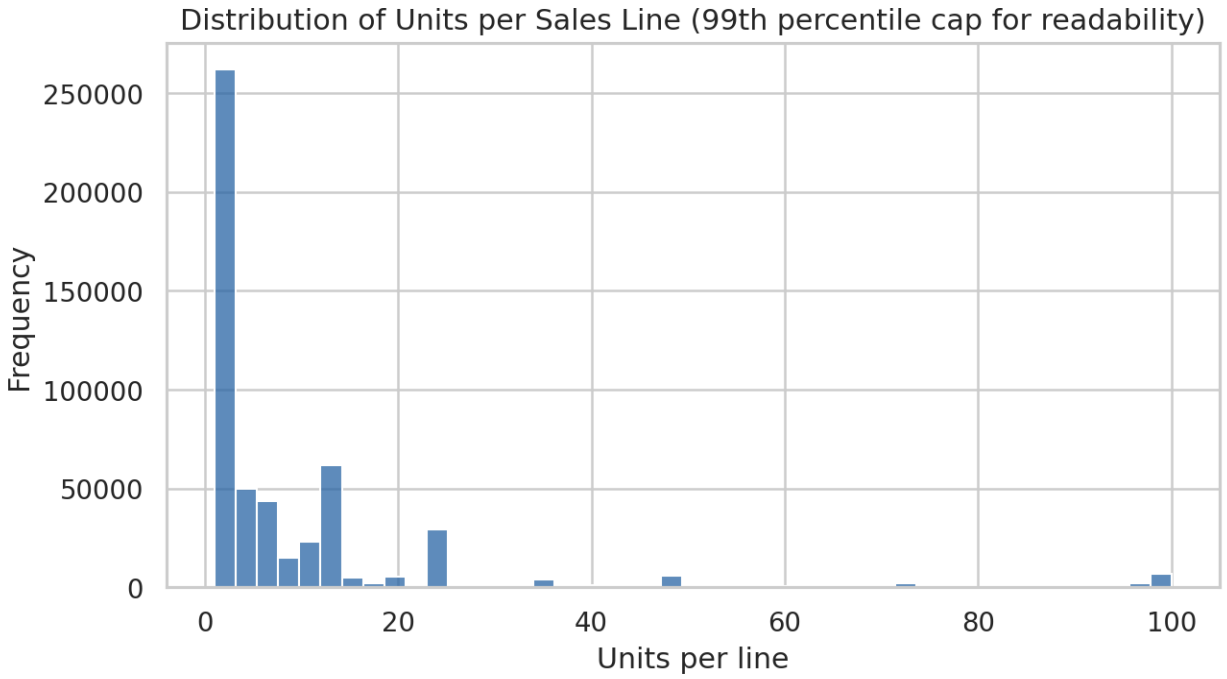


Figure 1. Quantity distribution. The visible bulk of sales lines contains relatively small unit quantities; a long upper tail indicates bulk purchases or exceptional lines.

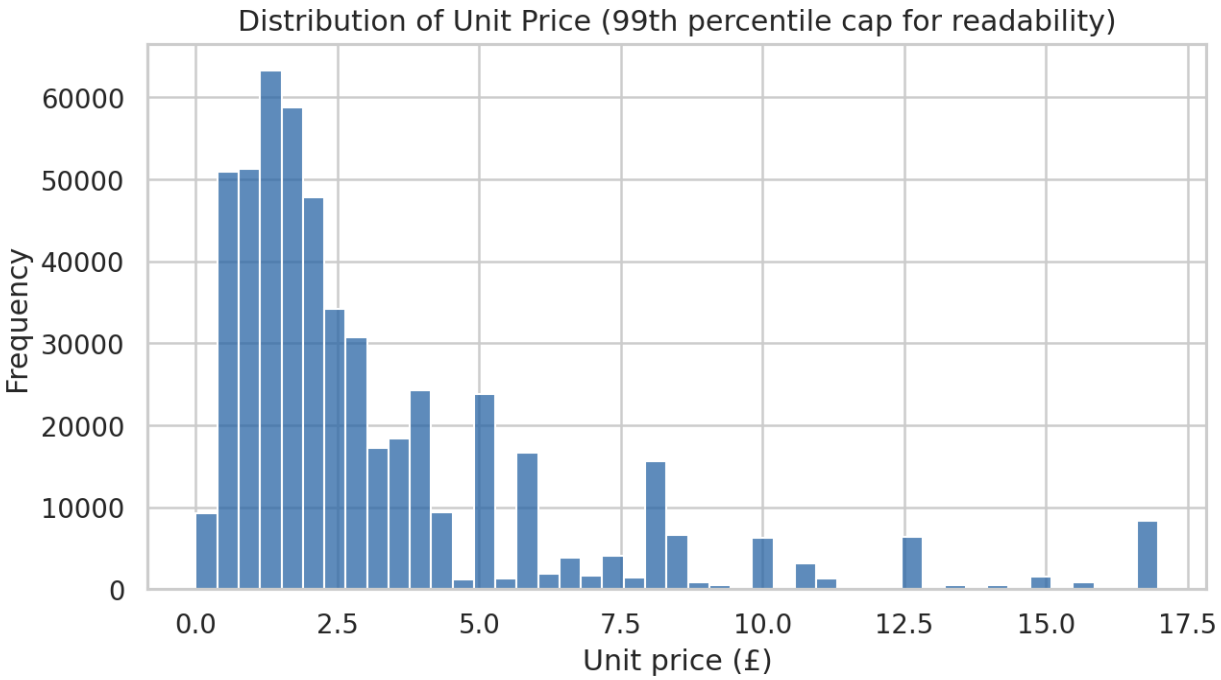


Figure 2. Unit-price distribution. Most line items are low-priced, with a small number of high-price items.

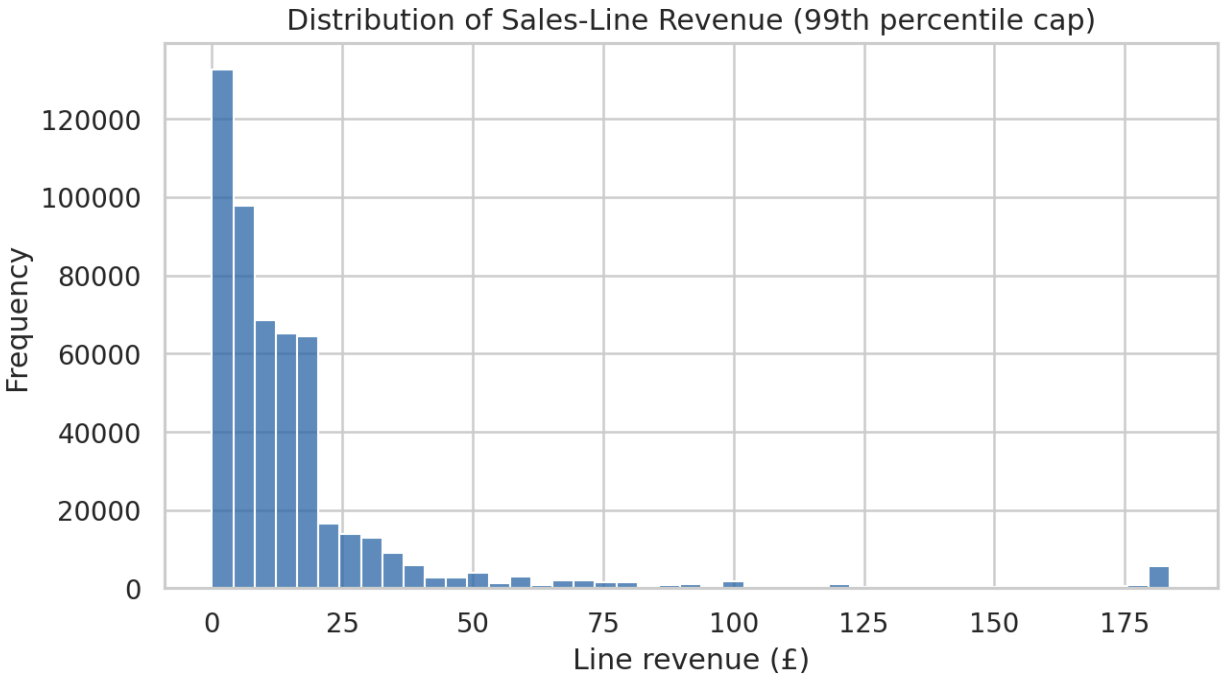


Figure 3. Line-revenue distribution. Revenue is concentrated in a large number of smaller line values with a long right tail.

5.2 Outlier review

The box plots reinforce the skew in quantity, unit price and line revenue. Extreme values should be investigated in context rather than automatically deleted: high quantities may reflect wholesale-like orders, and high unit prices may represent special items or non-standard stock codes.

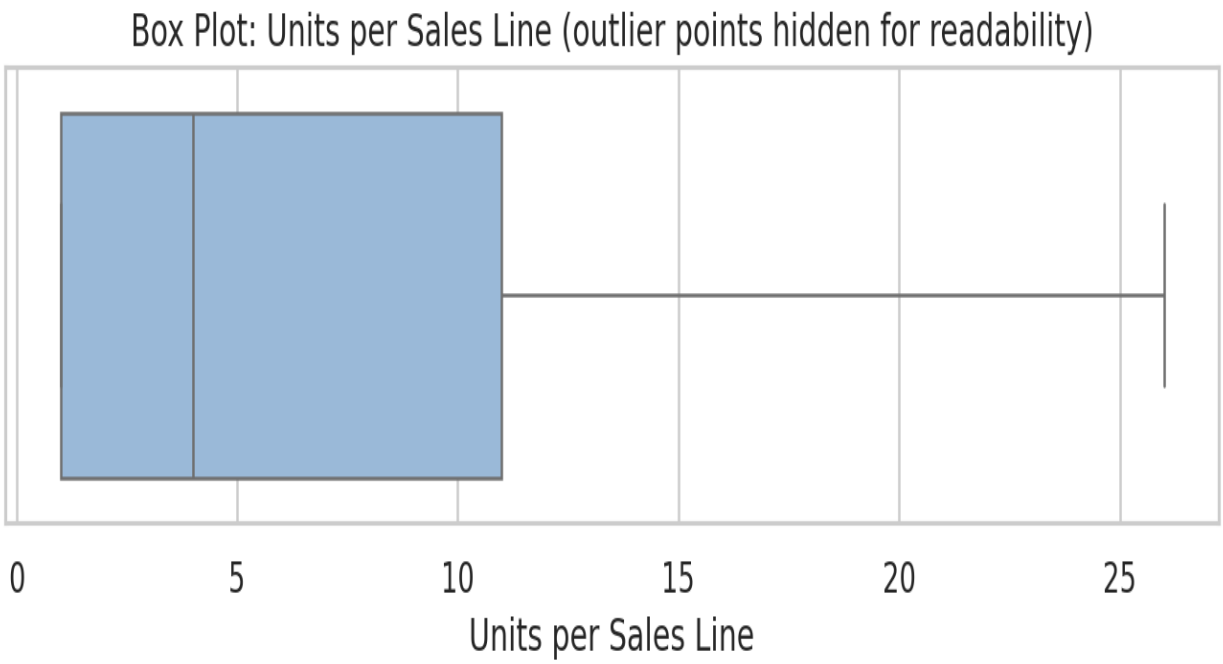


Figure 4. Quantity box plot; fliers hidden for readability.

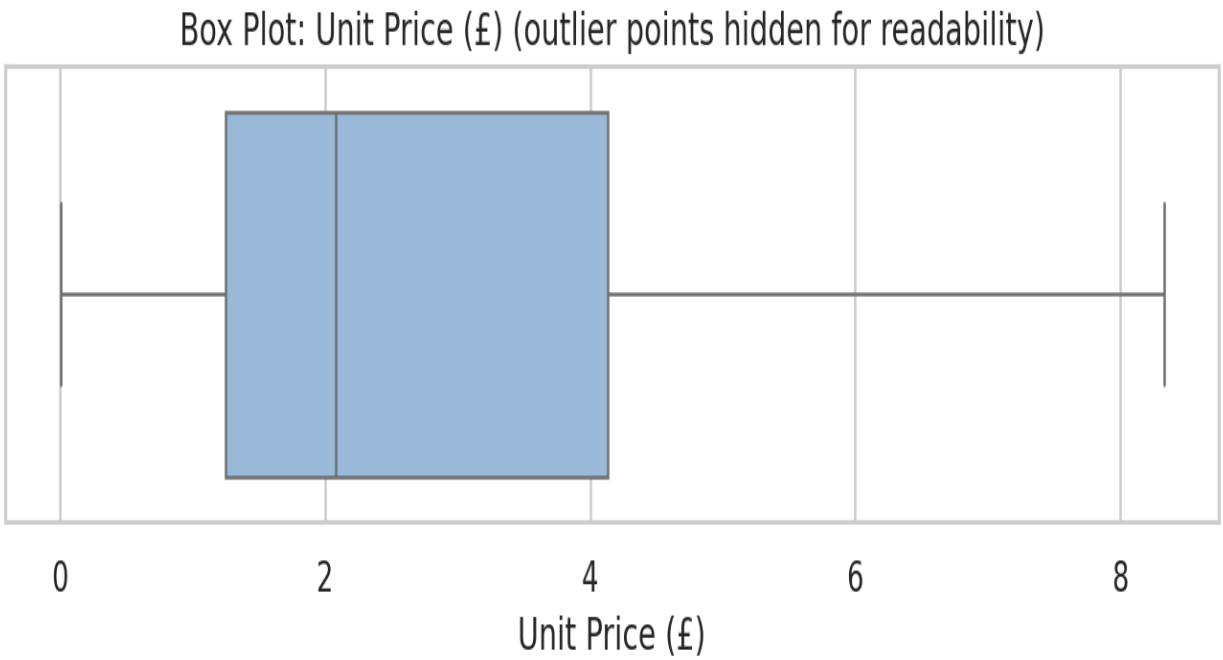


Figure 5. Unit-price box plot; fliers hidden for readability.

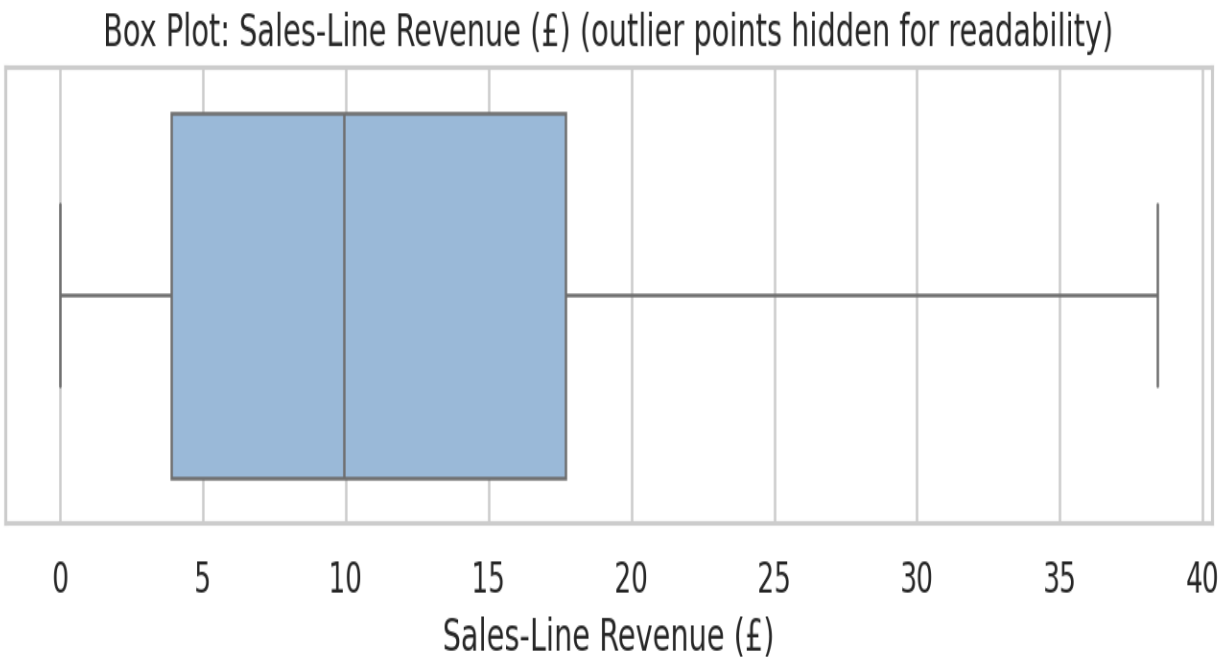


Figure 6. Line-revenue box plot; fliers hidden for readability.

5.3 Geographic and product performance

The country summary attributes £9,001,744 to the United Kingdom, equal to 84.6% of total revenue in the supplied KPI basis. The ten highest-revenue countries together contribute 97.2% of the sales-extract revenue. This indicates a highly concentrated geographic mix.

The leading product by revenue in the supplied product table is **DOTCOM POSTAGE** at £206,249. The leading product by units sold is **PAPER CRAFT , LITTLE BIRDIE** with 80,995 units. Revenue leadership and unit-volume leadership are not identical, so assortment decisions should consider both measures.

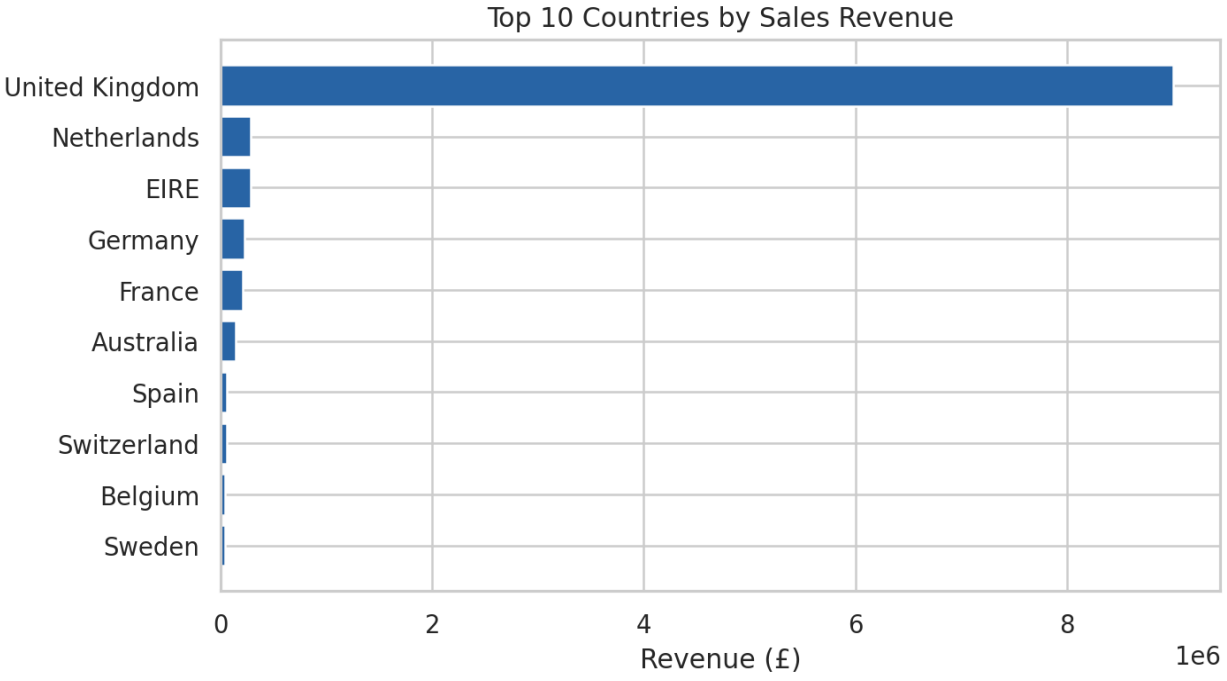


Figure 7. Top countries by revenue; the UK dominates the country summary.

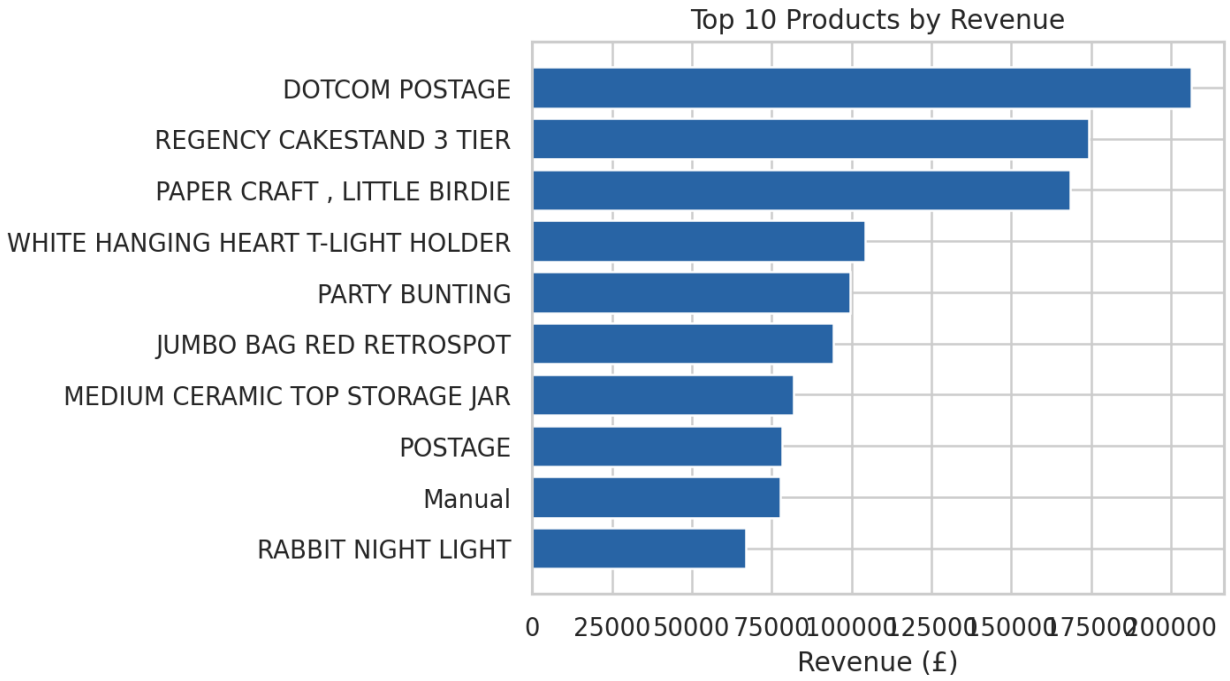


Figure 8. Top products by revenue; postage and several gift/home products feature prominently.

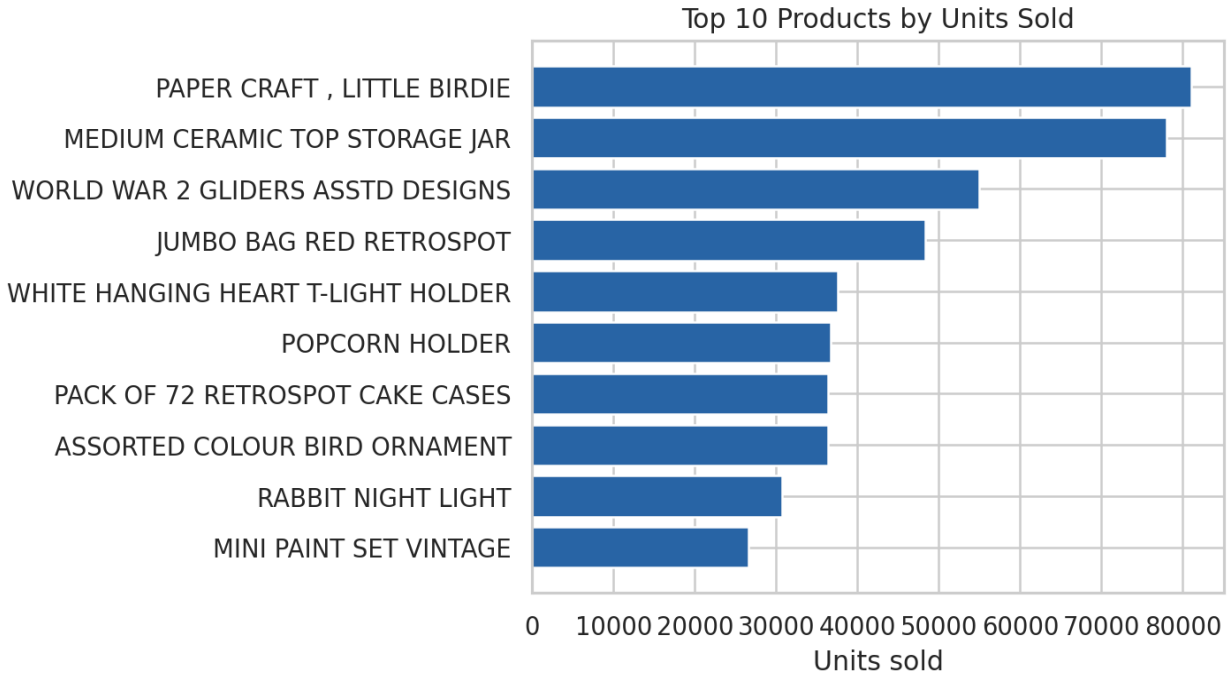


Figure 9. Top products by units sold; volume leaders differ from revenue leaders.

5.4 Time and transaction patterns

Monthly revenue reaches its maximum in November 2011 at £1,503,867. The supplied monthly series covers 13 month-start records from December 2010 through December 2011. December 2011 is a partial period in the sales extract, ending on 9 December, so its value should not be compared directly with complete months.

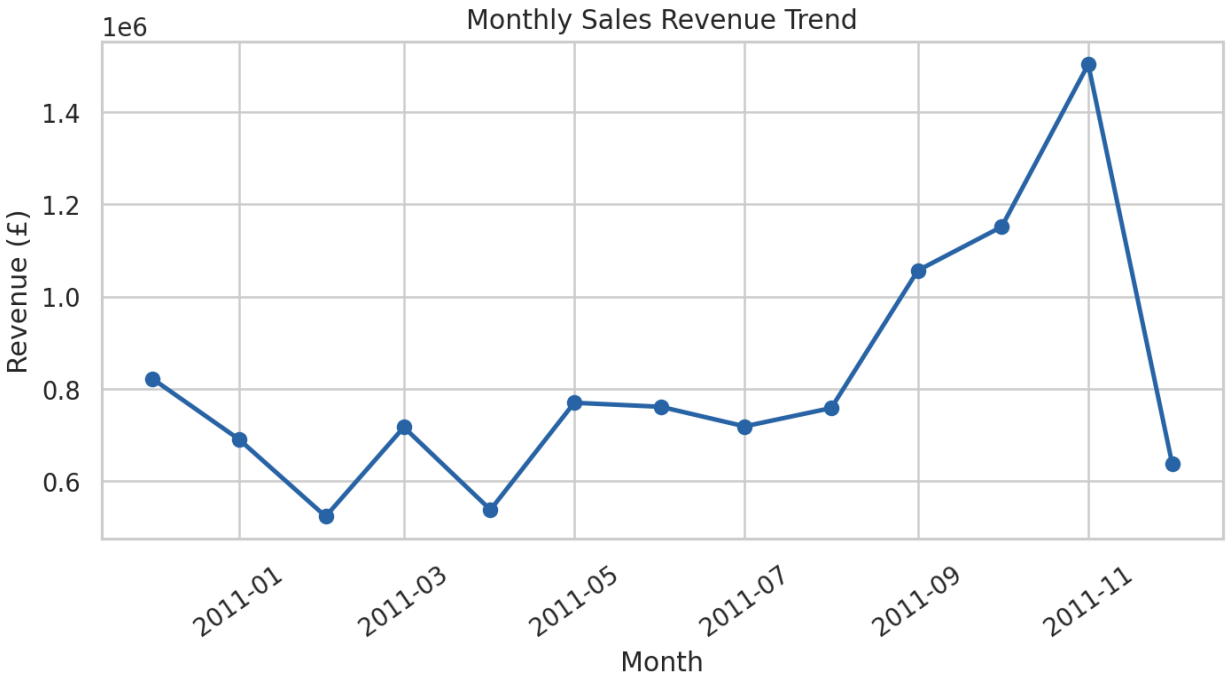


Figure 10. Monthly revenue trend. November 2011 is the peak; December 2011 is partial.

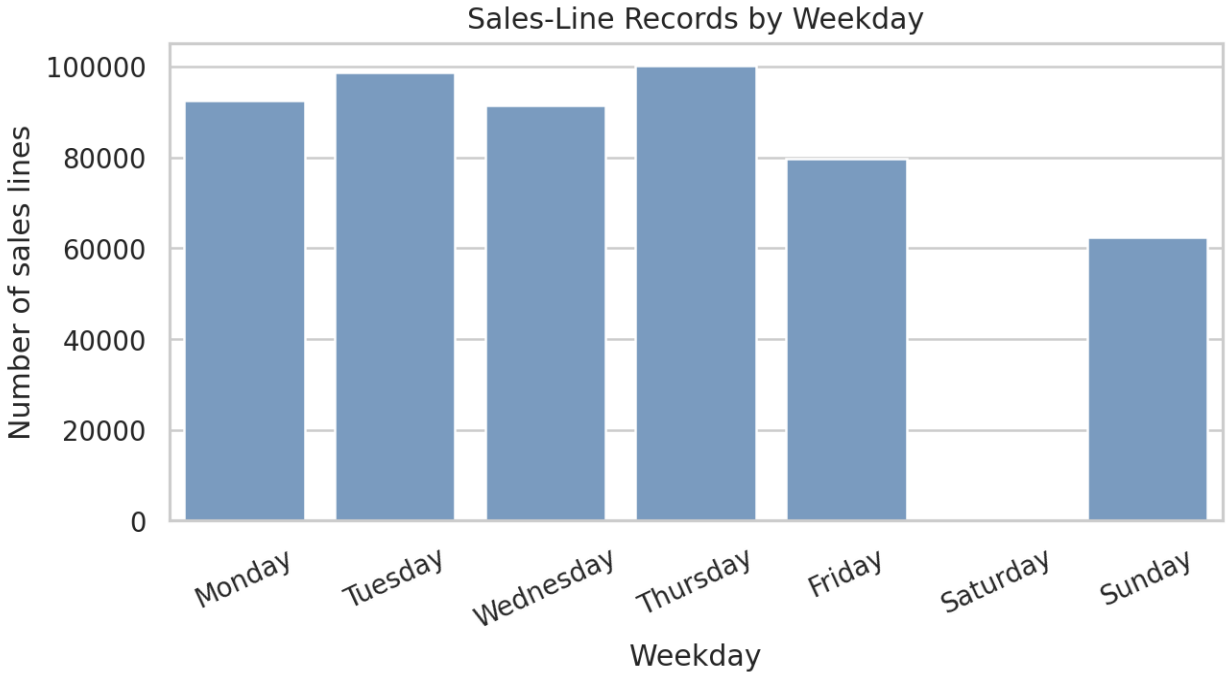


Figure 11. Sales-line record counts by weekday. No Saturday sales-line records appear in the supplied extract.

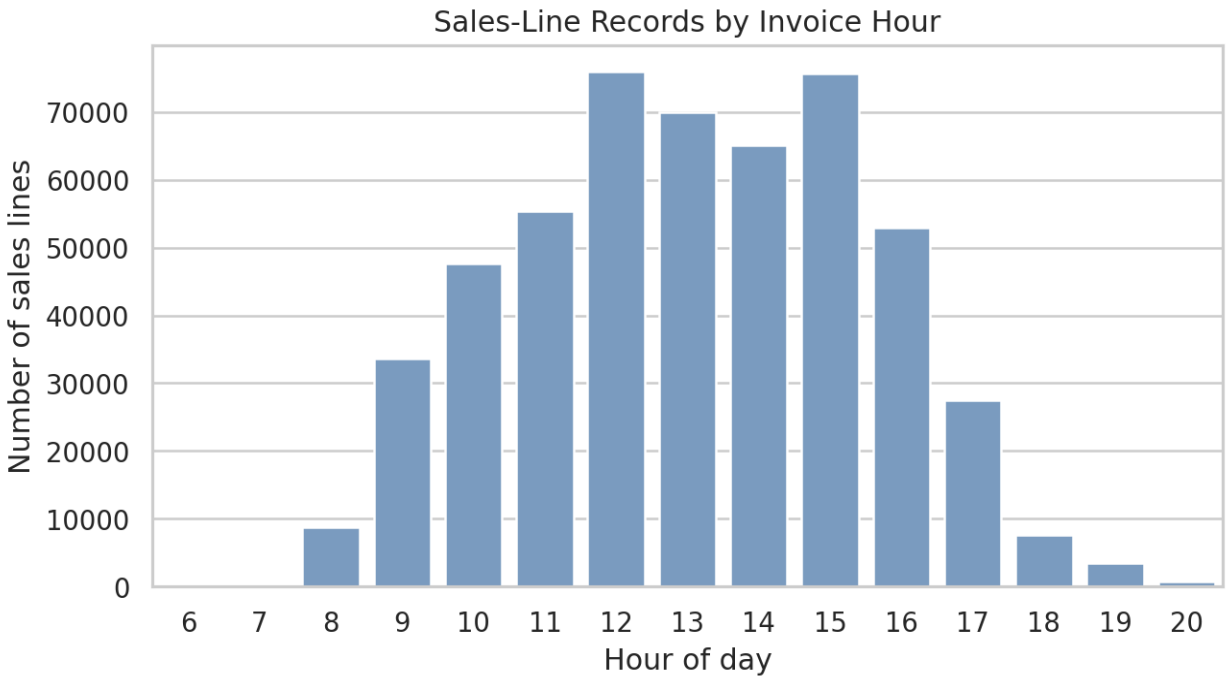


Figure 12. Sales-line record counts by invoice hour. Activity is concentrated in daytime trading hours.

5.5 Relationships and correlation

The scatter plots use logarithmic axes because both price/quantity and order-size/revenue measures are highly skewed. The relationship between unit price and quantity should not be interpreted as causal: product type, promotions, customer mix and order context may confound the pattern. Correlations are calculated at sales-line level and can be influenced by repeated products and outliers.

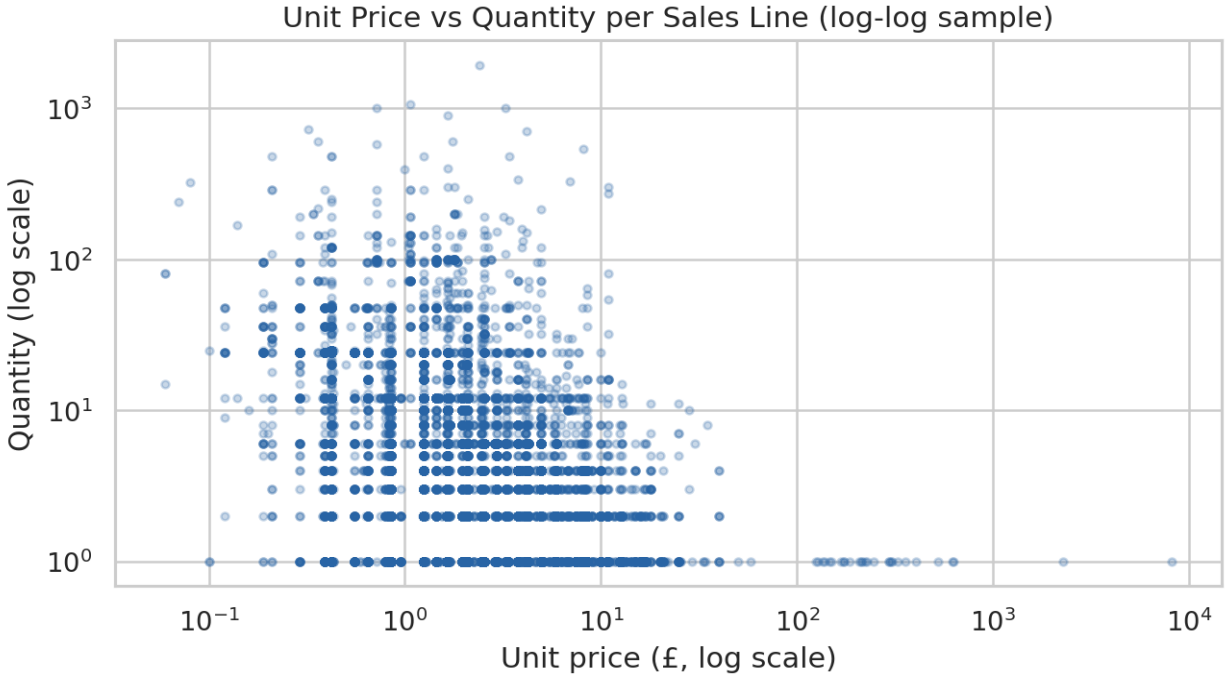


Figure 13. Unit price versus quantity per line, sampled for legibility and shown on log scales.

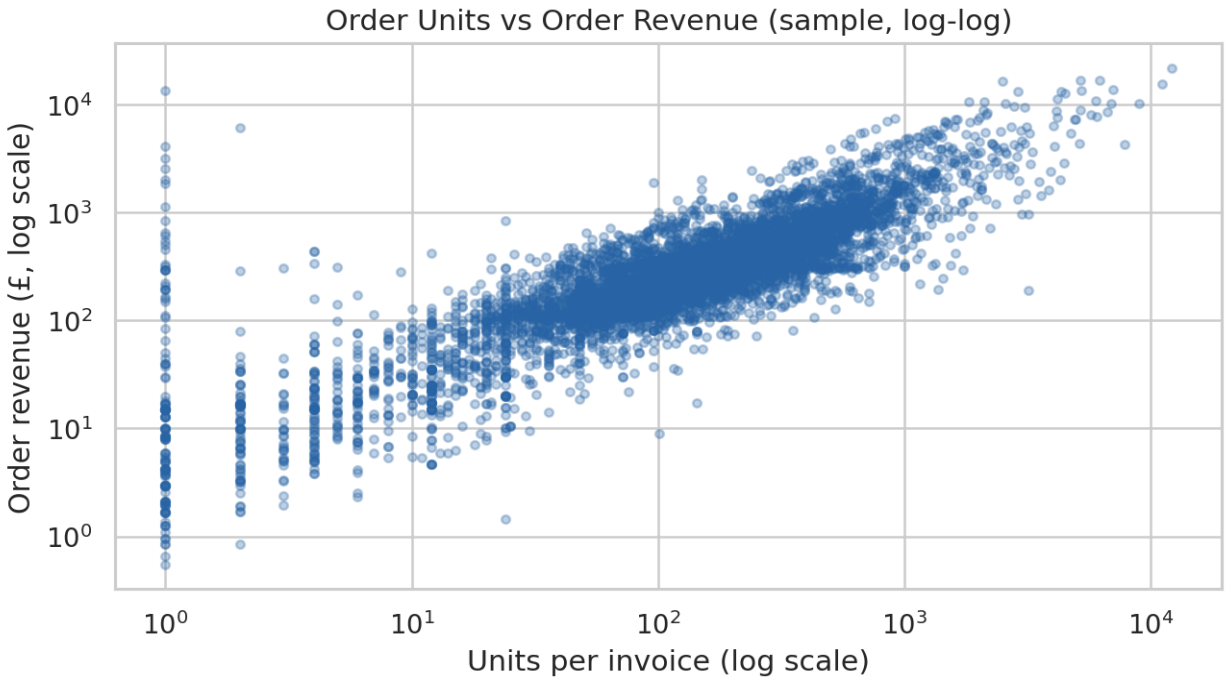


Figure 14. Order units versus order revenue; larger baskets generally have more opportunity to generate higher revenue, but variation remains.

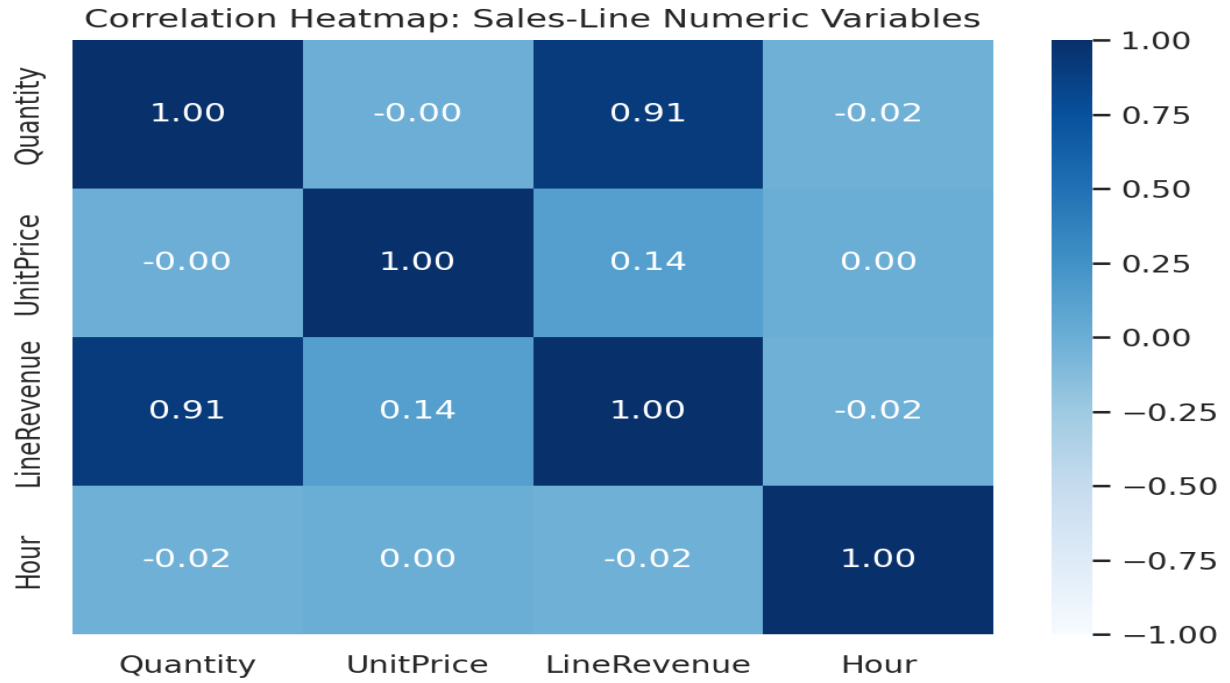


Figure 15. Correlation heatmap of quantity, unit price, line revenue and invoice hour. Correlation is descriptive, not causal.

6. Customer and Product Analysis

The KPI summary reports 4,338 identified customers. The supplied RFM summary segments these customers into six groups. Champions number 1,312, contribute £6,476,425 in segment revenue and have average recency of 17.0 days and average frequency of 9.4 invoices under the supplied RFM definitions.

The 'At Risk - Previously Active' segment contains 655 customers and £1.05m segment revenue. This segment is a reasonable candidate for carefully targeted re-engagement, subject to checking current contact permissions and campaign economics.

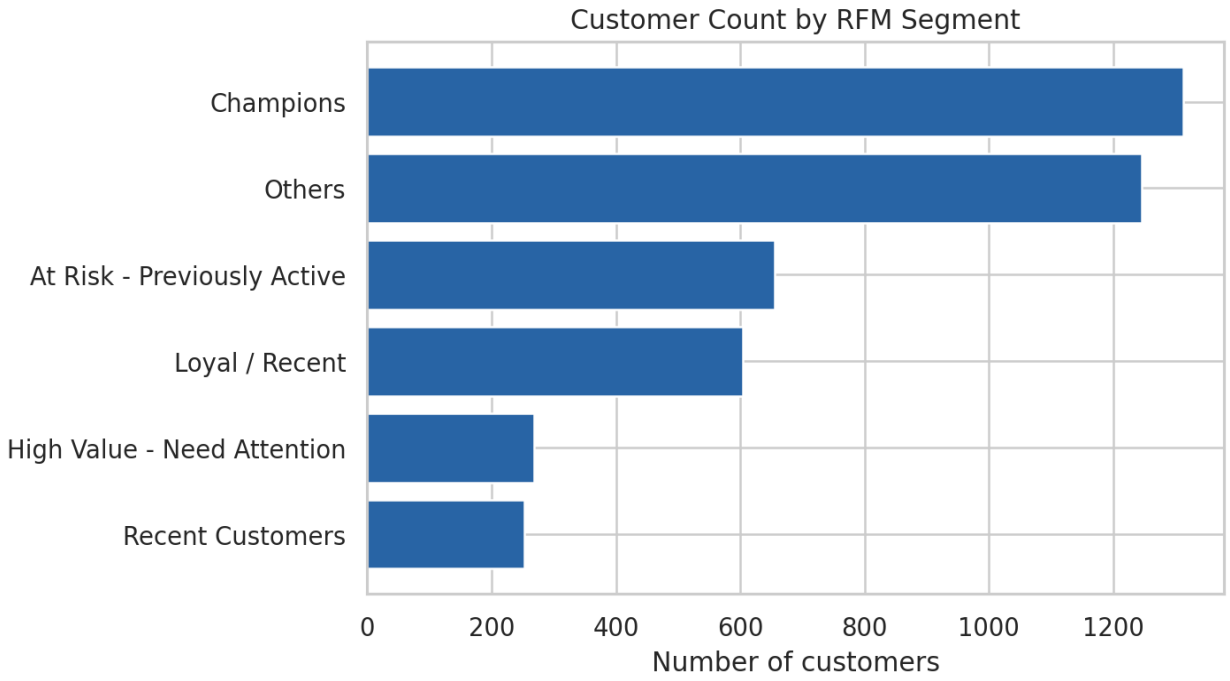


Figure 16. Customer counts by supplied RFM segment.

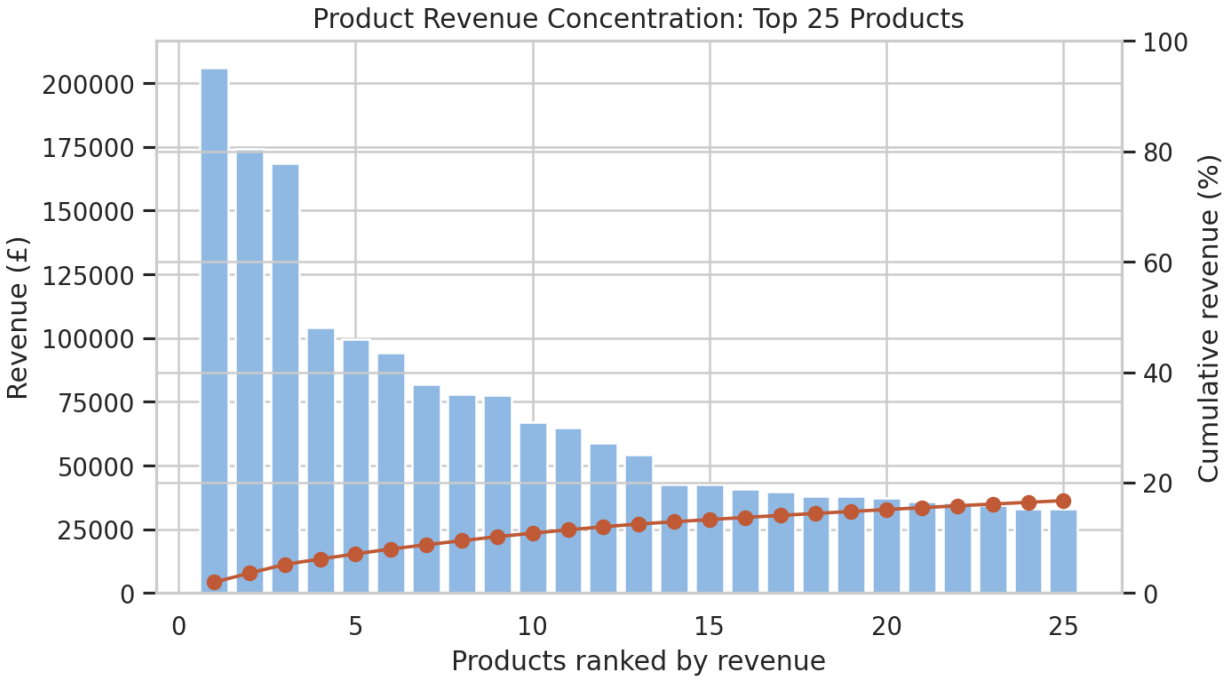


Figure 17. Revenue concentration across the top 25 products; the cumulative curve shows how revenue accumulates across ranked products.

The supplied KPI summary reports gross sales revenue of £10,642,110.80, 19,960 distinct invoices, 5,572,420 units sold, 4,338 identified customers, 3,922 unique product codes and average order value of £533.17. Independently aggregating the positive-sales extract yields 19,962 distinct invoice identifiers and a mean invoice revenue of £533.12. The small difference is disclosed rather than blended.

The positive-sales extract has 19,962 distinct invoice identifiers; aggregating its lines gives a median order revenue of £303.30. This median is materially below the mean, consistent with a right-skewed order-value distribution.

7. Key Findings and Recommendations

Geographic concentration

Observation: The UK contributes approximately 84.6% of revenue in the country summary.

Recommendation: Protect the core UK customer experience while testing international growth in a small number of proven markets.

Seasonal revenue peak

Observation: November 2011 is the highest-revenue month in the supplied monthly series (£1.50m).

Recommendation: Plan inventory, fulfilment capacity and customer support ahead of the peak season; validate with multiple years before treating it as a stable seasonal effect.

Customer value concentration

Observation: Champions represent 1,312 customers and about £6.48m in RFM segment revenue.

Recommendation: Use loyalty benefits, replenishment reminders and early access to retain valuable customers; monitor margin and incremental lift.

Reactivation opportunity

Observation: 655 customers are classified as At Risk - Previously Active with about £1.05m segment revenue.

Recommendation: Test a measured win-back campaign and compare conversion and incremental revenue against a holdout group.

Product mix differs by metric

Observation: The top revenue product and top volume product are different.

Recommendation: Use both revenue and units when planning assortment and replenishment; assess gross margin before prioritising products.

Data quality and reconciliation

Observation: CustomerID is missing on 25.2% of positive sales lines, and provided row/invoice totals do not fully reconcile.

Recommendation: Improve customer identification at checkout and validate source-to-output reconciliation before financial reporting.

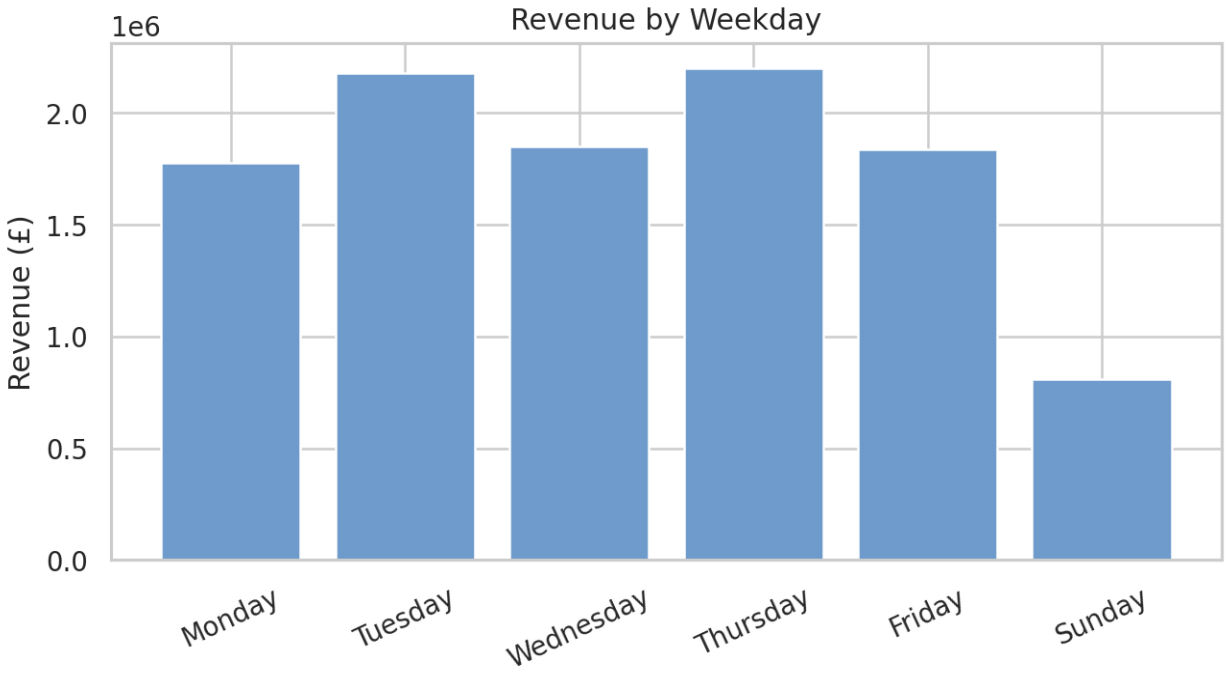


Figure 18. Revenue by weekday; use alongside staffing and fulfilment capacity planning, while accounting for the dataset's trading-day coverage.

8. Limitations

• This report analyses the supplied prepared extracts and summary tables, not a freshly downloaded original UCI workbook.
• The original raw workbook and preprocessing script were not included in the ZIP, so exact reconciliation cannot be completed here.
• The KPI summary and transaction extract differ by two invoice identifiers; cleaned-master row counts also differ from the two supplied extracts by 1,176 rows.
• CustomerID is missing for 25.2% of positive sales lines, limiting customer-level coverage.
• The period covers approximately one year, with December 2011 ending on 9 December; this limits robust year-over-year seasonality conclusions.
• RFM segments depend on the segmentation method and reference date used in the supplied outputs; those parameters should be documented in the source notebook.
• Descriptive EDA cannot establish causal effects. No profitability, marketing cost, stock availability or customer lifetime value data were supplied.

9. Conclusion

The supplied Online Retail outputs show a large sales base, strong UK revenue concentration, a pronounced November 2011 peak, a distinct group of high-value Champions and a product mix where revenue and volume leaders differ. These patterns can guide customer retention, reactivation, inventory planning and geographic strategy. The project also surfaces material data-quality caveats—especially missing customer identifiers and reconciliation differences—that should be resolved before using the outputs for audited financial decisions.

10. References

UCI Machine Learning Repository, Online Retail Dataset: https://archive.ics.uci.edu/dataset/352/online+retail
Assignment brief: EDA Projects Bleep, pages 2 and 7–11 (project questions, workflow, visualization minimums and submission requirements).
Python libraries used: Pandas, NumPy, Matplotlib and Seaborn.

Appendix: Supplied KPI Summary

Metric	Value
Gross sales revenue	10,642,110.80
Distinct invoices	19,960.00
Units sold	5,572,420.00
Unique identified customers	4,338.00
Unique product codes	3,922.00
Average order value	533.17
Average units per invoice	279.18